Bucket Sort and Large-Scale Problem Decomposition

Design and Analysis of Algorithms
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Warm-Up: Balls in Bins Revisited

- Throw N balls randomly into M bins.
- Already discussed: **E**[balls per bin] = N / M.
- How about **E**[number of pairs of balls that "collide", landing in the same bin]? $N(N - 1) / (2M) \approx N^2 / (2M)$
  - Number of pairs: $\binom{N}{2} = N(N - 1) / 2$
  - Probability a pair collides: 1 / M

  E[total collisions] = $\sum_{i<j}$ **E**[X_{ij}] (where $X_{ij} = 1$ if balls i & j collide)

  If N = M, **E**[total collisions] = $(N - 1) / 2 \approx N / 2$

Related: The Birthday "Paradox"

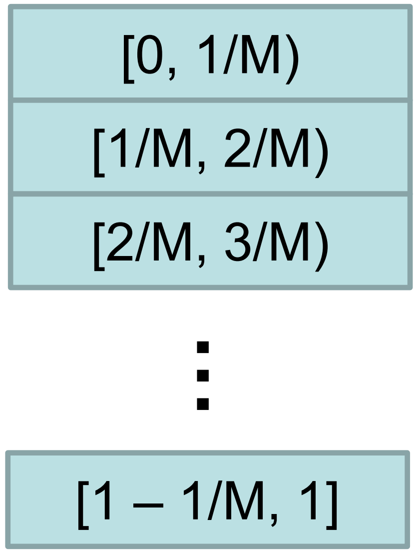
- N people in a room.
- How large can N get before you anticipate seeing a shared birthday? (assuming all birthdays equally likely)
- **E**[# collisions for N balls in 365 bins] ≈ N^2 / (2 x 365)
  - If N is $0.1\sqrt{365}$, this is 0.005
  - If N is $\sqrt{365}$, this is 0.5
  - If N is $10\sqrt{365}$, this is 50
- Ballpark threshold: $\sqrt{(\#\ \text{days})}$

Birthday "Paradox" Applications

- Sample up to $\sqrt{N}$ elements from pool of size N
  - Reasonable chance they're all distinct
- Sample elements from large pool, stopping when the Nth element is one you've seen before
  - Estimate $\Theta(N^2)$ as pool size
- Map up to $\sqrt{N}$ elements into a hash table of size N
  - Reasonable chance they all end up in distinct table cells
- Store copies of object on $\sqrt{N}$ random servers (of N total)
  - Reasonable chance of finding if probe $\sqrt{N}$ random servers

Bucket Sort: Sorts Samples from Known Distribution in O(N) Expected Time

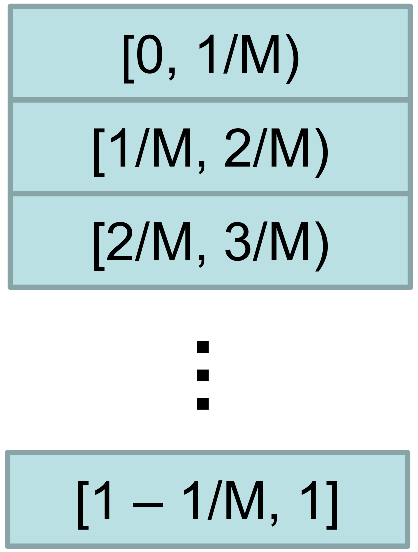
- Sorting N numbers uniformly drawn from [0, 1].
- We could run bubble sort in $O(N^2)$ time.
- Or, spread into M buckets (in O(N) time), then bubble sort the N / M expected elements in each.
- "Back of the envelope" math:
  - $O((N / M)^2)$ time / bucket times M buckets: $O(N^2 / M)$.
  - Factor of M speedup! (in particular, just O(N) if we set M = N).
  - This is the right asymptotic answer in expectation, but we need to be a bit more careful with the math…



Problem Decomposition into Multiple Pieces

- Can often solve problems more efficiently by dividing them into a large number of pieces that are either independent or minimally-dependent.

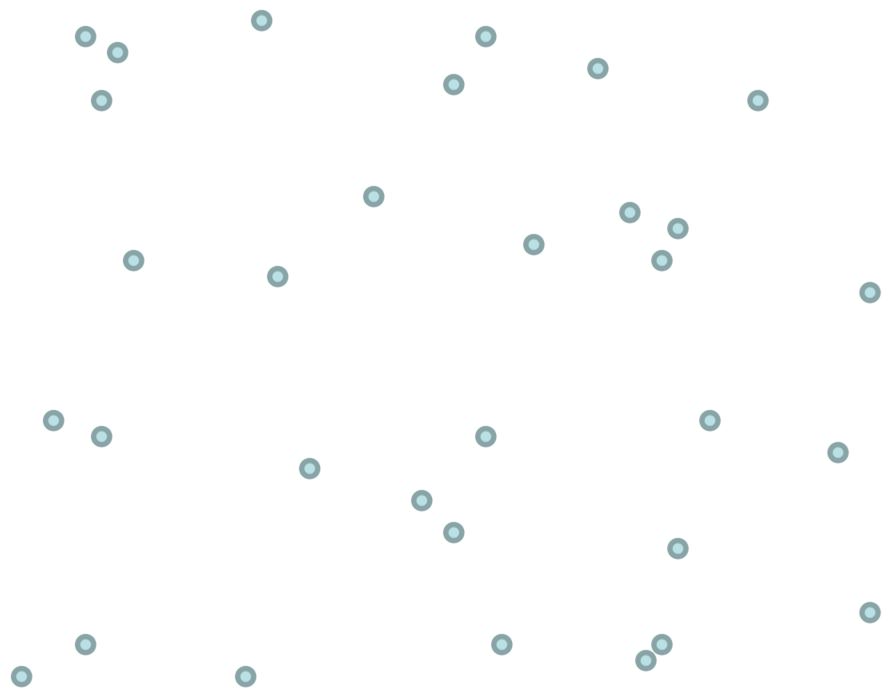
  (e.g., N^2 vs. $M \times (N / M)^2 = N^2 / M$)



- Related to "divide and conquer" design approach ("divide and conquer" usually recurses on the pieces, further subdividing them).

Closest Pair of Points

Given N points uniformly spread across the unit square in 2D, find the closest pair.



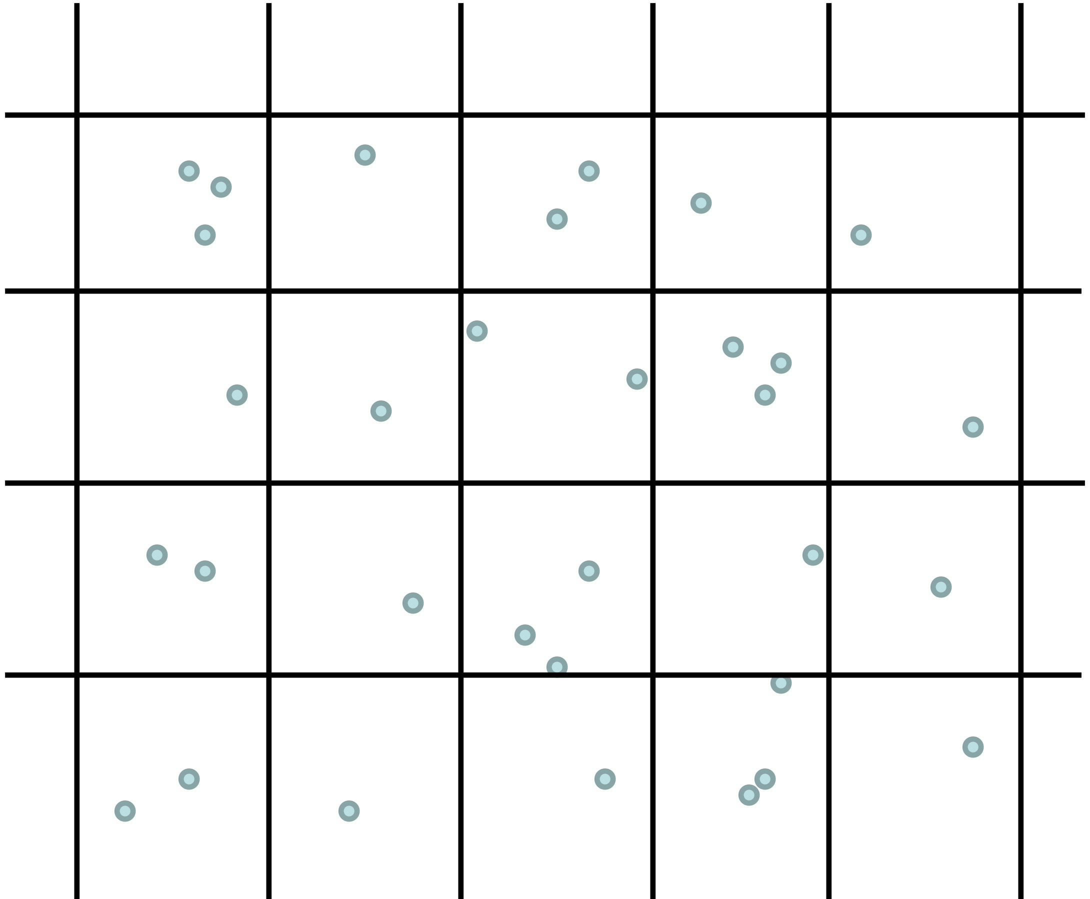
Closest Pair of Points

Given N points uniformly spread across the unit square in 2D, find the closest pair.

Use $\sqrt{N} \times \sqrt{N}$ grid to make N "bins"

So N balls → N bins

E[# balls / bin] = 1



1. Compare all pairs of points: $\Theta(N^2)$.

2. Compare each point only to points in its bin (and surrounding bins): $\Theta(N)$ in expectation!